Knowledge Graphs

Lecture 4 - Knowledge Representation with Ontologies

4.3 - First Steps in OWL

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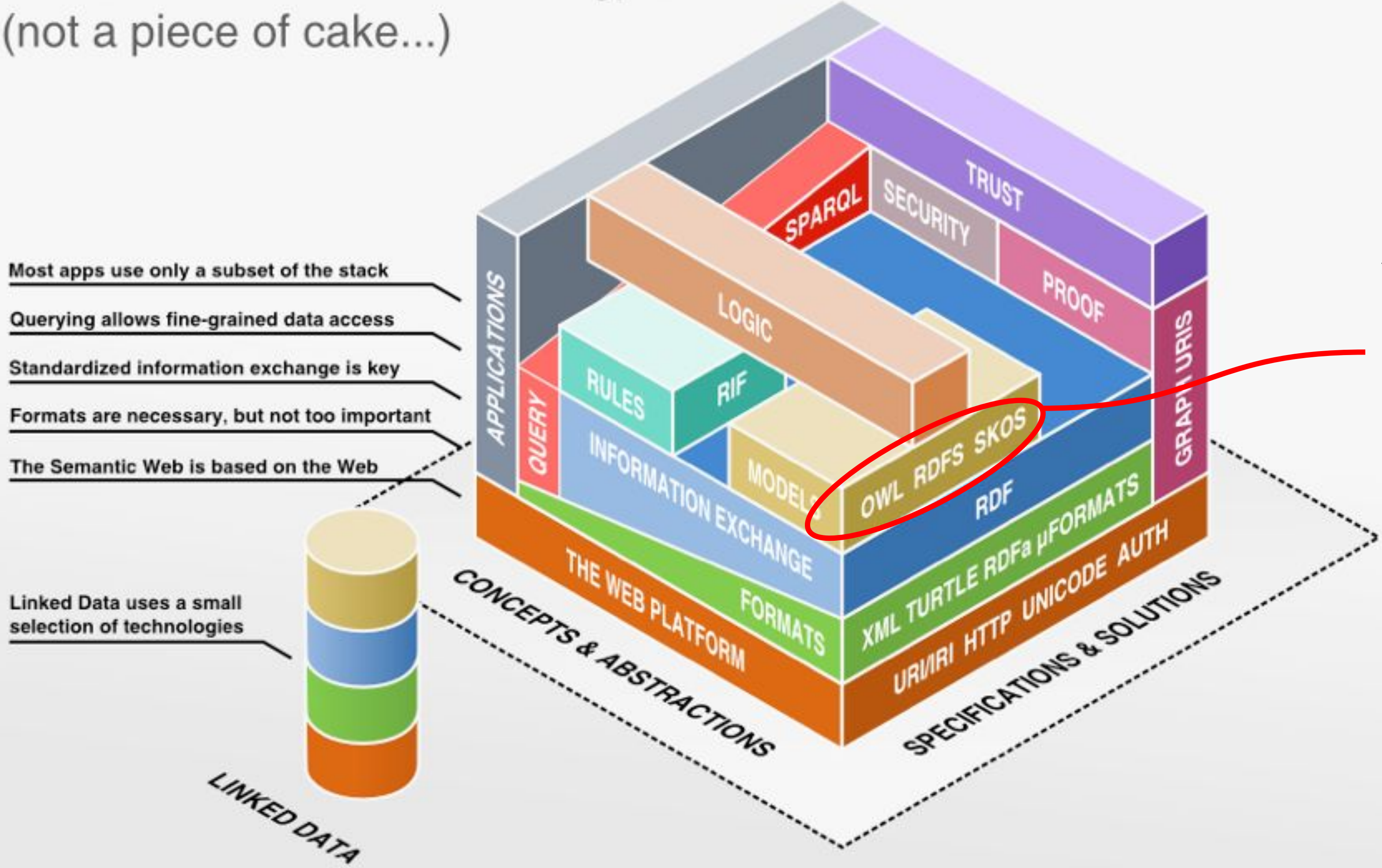
Knowledge Graphs

Lecture 4: Knowledge Representation with Ontologies

The Semantic Web Technology Stack
(not a piece of cake...)



Web
Ontology
Language
(OWL)

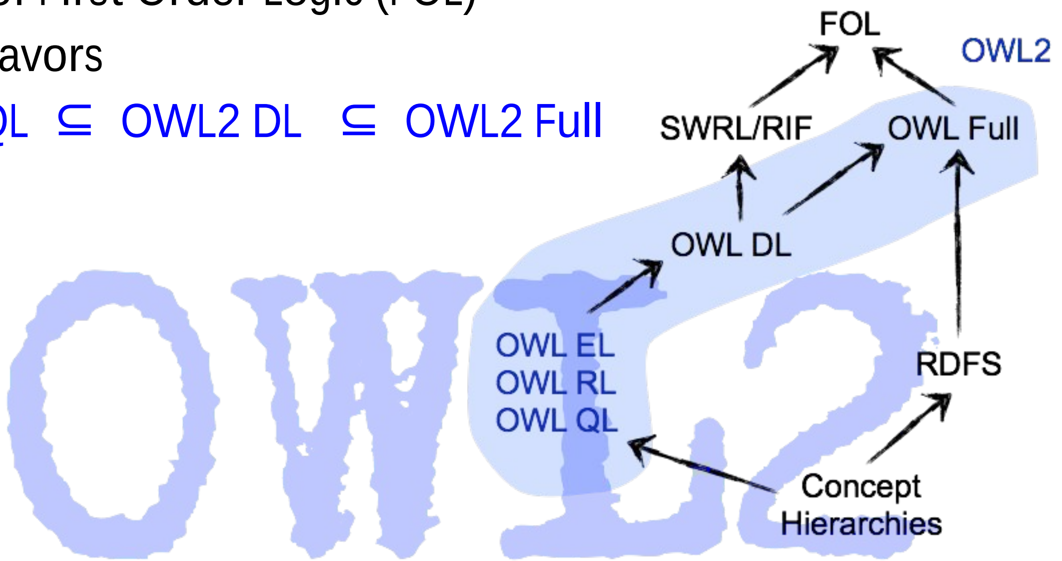
[1]

Web Ontology Language OWL - OWL Flavours

OWL is a semantic fragment of First Order Logic (FOL)

OWL also exists in different flavors

- OWL EL, OWL RL, OWL QL $\subseteq$ OWL2 DL $\subseteq$ OWL2 Full



OWL2 is based on the Description Logic *SROIQ(D)*

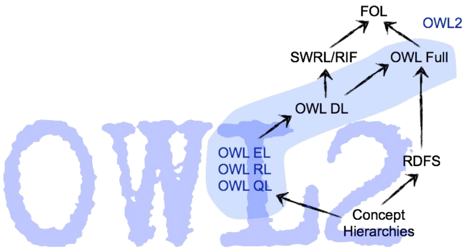
Class Expressions
- Class names A,B
- Conjunction $C \sqcap D$
- Disjunction $C \sqcup D$
- Negation $\neg C$
- Exist. property restriction $\exists R.C$
- Univ. property restriction $\forall R.C$
- Self $\exists S.Self$
- Greater-than n S.C
- Less-than n S.C
- Enumerated classes {a}

Properties
- Property names R,S,T
- Simple properties S,T
- Inverse properties R^-
- Universal property U

Tbox (Class axioms)
- Inclusion $C \sqsubseteq D$
- Equivalence $C \equiv D$

Rbox (Property Axioms)
- Inclusion $R_1 \sqsubseteq R_2$
- General Inclusion $R^{(-)}_1 \circ R^{(-)}_2 \circ \dots \circ R^{(-)}_n \sqsubseteq R$
- Transitivity
- Symmetry
- Reflexivity
- Irreflexivity
- Disjunctiveness

Abox (Facts)
- Class membership C(a)
- Property relation R(a,b)
- Negated property relation $\neg S(a,b)$
- Equality a=b
- Inequality a b



OWL Basic Building Blocks

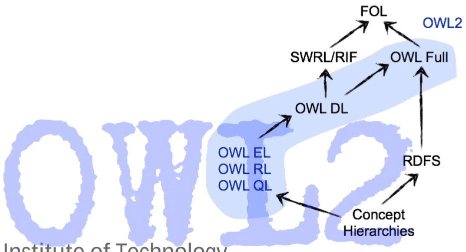
OWL namespace:

```
@prefix owl: <http://www.w3.org/2002/07/owl#>
```

There is a Turtle Syntax for OWL

OWL axioms consist of the following three building blocks:

- Classes
  - comparable with classes in RDFS
- Individuals
  - comparable with class instances in RDFS
- Properties
  - comparable with properties in RDFS



OWL Classes

- There exist two predefined classes
  - `owl:Thing` (class that contains all individuals)
  - `owl:Nothing` (empty class)

equivalent expression in description logics

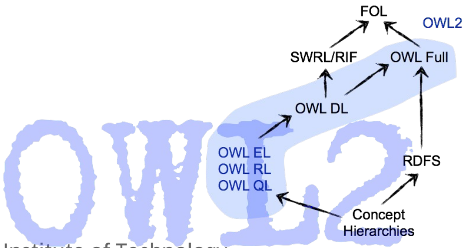
$\top \quad C \sqcup \neg C$

$\bot \quad C \sqcap \neg C$

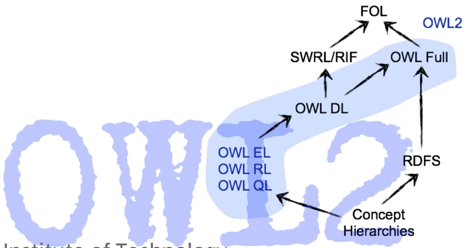
- Definition of a class

```
:GreenhouseGas a owl:Class .
```

This is OWL in RDF/Turtle serialization



OWL Individuals



OWL Datatype Properties

Datatype properties have datatypes as range

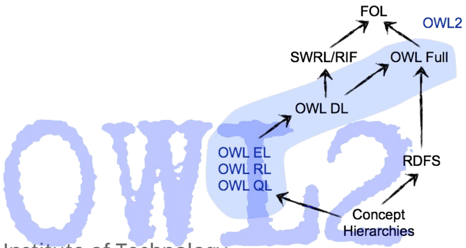
```
:discoveredIn a owl:DatatypeProperty .
```

Domain and Range of datatype properties

```
:discoveredIn a owl:DatatypeProperty ;
              rdfs:domain owl:Thing ;
              rdfs:range  xsd:date .
```

$\exists \mathtt{discoveredIn}.\top \sqsubseteq \top$

$\top \sqsubseteq \forall \mathtt{discoveredIn}.\mathtt{Date}$



OWL Properties and Individuals

OWL TBox

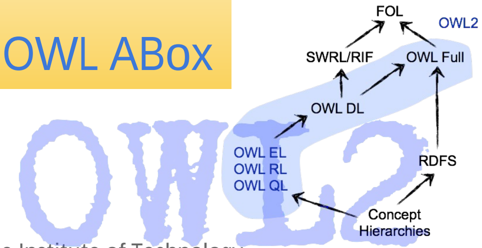
```
:AtmosphericProcess a owl:Class .
:Person a owl:Class .

:discoverer a owl:ObjectProperty ;
            rdfs:domain owl:Thing ;
            rdfs:range  :Person .

:discoveredIn a owl:DatatypeProperty ;
              rdfs:domain owl:Thing ;
              rdfs:range  xsd:date .
```

```
:JosephFourier a Person .
:GreenhouseEffect a :AtmosphericProcess ;
               :discoverer :JosephFourier ;
               :discovered “1824-00-00”^^xsd:date .
```

OWL ABox



OWL Class Hierarchies

```
:Physicist a owl:Class ;
       rdfs:subClassOf :Scientist .
:Scientist a owl:Class ;
      rdfs:subClassOf :Person .
:Person a owl:Class .
```

we don't need to define a new subClassOf property for OWL, we simply reuse `rdfs:subClassOf`

Physicist ⊑ Scientist
Scientist ⊑ Person

via inference it can be entailed that `:Physicist` is also a subclass of `:Person`

OWL Class Hierarchies and Disjunctiveness

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```
:ChemicalSubstance a owl:Class .
:Person a owl:Class .
:GreenhouseGas a owl:Class ;
       rdfs:subClassOf :ChemicalSubstance .
:Scientist a owl:Class ;
      rdfs:subClassOf :Person .

:ChemicalSubstance owl:disjointWith :Person .
```

In OWL everything might be potentially identical if we don't explicitly state the difference

$$\mathrm{GreenhouseGas} \sqsubseteq \mathrm{ChemicalSubstance}$$
$$\mathrm{Scientist} \sqsubseteq \mathrm{Person}$$
$$\mathrm{ChemicalSubstance} \sqcap \mathrm{Person} \sqsubseteq \bot$$

via inference it can be entailed that `:GreenhouseGas` and `:Scientist` are also disjoint classes

OWL Class Hierarchies and Equivalence

```
:Scientist  a owl:Class .
:Researcher a owl:Class .
:Physicist a owl:Class ;
      rdfs:subClassOf :Scientist .

:Scientist owl:equivalentClass :Researcher .
```

$\text{Physicist} \sqsubseteq \text{Scientist}$

Scientist   Researcher

via inference it can be entailed that `:Physicist` is also a `:Researcher`

OWL Individuals - Identity and Distinctiveness

```
:CarbonDioxide a :GreenhouseGas ;
    :discoverer :JosephBlack ;
    :discoveredIn “1750-00-00”^^xsd:date ;
    owl:sameAs :ARX012345 .

:GreenhouseGas a owl:Class ;
       rdfs:subClassOf :ChemicalSubstance .

:ChemicalSubstance a owl:Class.
```

For identical individuals: `owl:sameAs`
For identical classes: `owl:equivalentClass`

via inference it can be entailed that `:ARX012345` is a `:ChemicalSubstance`

difference of Individuals via `owl:differentFrom`

```
:ARX012345 a :GreenhouseGas ;
           owl:differentFrom :ARX012346 .
```


More OWL
Next Lecture...

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Picture References:

[1] Benjamin Nowack, *The Semantic Web - Not a Piece of cake…*, at bnode.org, 2009-07-08 , [CC BY 3.0]
http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake

[2] Gustave Doré, Two Owls, 19th century [Public Domain]
https://www.wikiart.org/en/gustave-dore/two-owls